INNOVATIVE SIMULATION WORKFLOW FOR ELECTRO-THERMAL SIMULATION OF PHASED-ARRAY ANTENNAS

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***Abstract*—Active electronically scanned arrays (AESAs) are increasingly adopted in space, defense, and telecommunication systems due to their beam-steering agility and high aperture efficiency. However, the dense integration of radiating elements, transmit/receive modules, and high-power amplification stages introduces significant thermal loads, affecting electromagnetic performance, reliability, and design margins. In practical workflows, electromagnetic and thermal analyses remain partially disconnected, requiring manual geometry reconstruction, loss transfer, and temperature reassignment. This paper presents a scripted electromagnetic-to-thermal co-simulation workflow that preserves the efficiency of unit-cell and finite-array electromagnetic modeling while enabling a closed, repeatable electro-thermal analysis at aperture level, suitable for early design iterations.**

I. Introduction

Through the last two decades, active electronically scanned arrays (AESAs) have been widely adopted in space, defense, radar, and telecommunication systems thanks to rapid beam steering and flexible control of the radiated field. Their performance hinges on the stability of amplitude and phase excitations across densely integrated radiating elements and transmit/receive modules (TRMs) [1], [8]. As array size, output power, and integration density grow, thermal effects degrade gain, sidelobe level, pointing accuracy, and taper effectiveness [1], [3], with additional geometrical phase errors from thermal deformation [5], [6]; the issue is particularly severe in spaceborne systems where mass, packaging, and cooling are constrained.

Over the past years, studies have addressed individual facets of the coupled problem—temperature-dependent EM degradation, thermal-aware synthesis, electro-thermal TRM modeling, and multiphysics analysis [1], [2], [4], [5], [6], [7]. However, full-wave simulation of a large AESA without domain-decomposition techniques such as FADDM is computationally impractical, and transferring EM losses from a unit-cell model to an aperture-level thermal model still requires manual geometry reconstruction, power scaling, source assignment, and setup replication—making design-space exploration slow and poorly repeatable.

Motivated by these limitations, this paper presents an automated HFSS-to-Icepak coupling procedure. Losses extracted from an HFSS unit-cell solution are transferred to an aperture-level thermal model while preserving their association with physical components and accounting for the beamforming taper. The Icepak model is not constrained to the HFSS unit-cell geometry, enabling non-radiating components (mechanical supports, enclosures, cooling structures) and thermal analyses ranging from preliminary conduction to detailed cooling and radiative studies, with automatic per-cell hotspot indicators.

Yet, structural analysis remains outside the scope of this work. Evaluating thermally induced aperture deformation requires a dedicated mechanical model and is left to future extensions of the proposed workflow.

More specifically, these challenges are illustrated throughout this paper on a representative 18×18 AESA test case (Fig. 1), used as a running example to quantify the impact of thermal loading on gain, beamwidth, and pointing accuracy.

Using the array far-field formalism, the electromagnetic performance of an AESA is governed by the complex excitation distribution across its aperture. For a planar array of $M \times N$ elements, the radiated field can be expressed as

$$E(\theta,\phi,T) = \sum_{m=1}^{M}\sum_{n=1}^{N} A_{mn}(T)\, f_{mn}(\theta,\phi) \times \exp\left\{j\left[k\mathbf{r}_{mn}\cdot\hat{\mathbf{u}}(\theta,\phi)+\phi_{mn}(T)\right]\right\}, \tag{1}$$

where $A_{mn}(T)$ and $\phi_{mn}(T)$ are the temperature-dependent excitation amplitude and phase, $f_{mn}(\theta,\phi)$ is the element radiation pattern, $\mathbf{r}_{mn}$ is the element position vector, $\hat{\mathbf{u}}(\theta,\phi)$ is the observation-direction unit vector, and $k=2\pi/\lambda$ is the free-space wavenumber at the operating frequency.

Moreover, operating AESAs depart from their nominal excitation state because the active RF chain is temperature dependent, so that $A_{mn}(T)=A^0_{mn}+\Delta A_{mn}(T)$ and $\phi_{mn}(T)=\phi^0_{mn}+\Delta\phi_{mn}(T)$, where A^0_{mn}, ϕ^0_{mn}

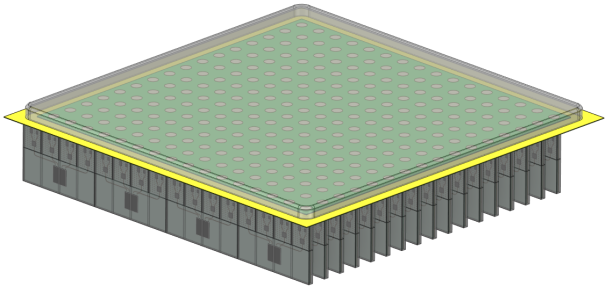


Fig. 1: AESA test case geometry: 18×18 planar array (central active aperture surrounded by a one-cell passive frame) with the back-side TRM chain.

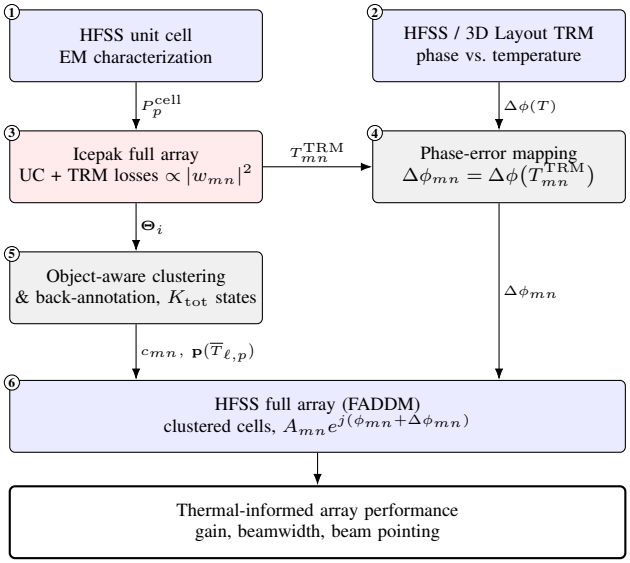


Fig. 2: Thermal-informed array analysis workflow (blue: EM; red: thermal; gray: post-processing).

follow the beam-steering law and taper, and the perturbations $\Delta A_{mn}(T)$, $\Delta\phi_{mn}(T)$ accumulate across the aperture, degrading gain, sidelobe level, and pointing accuracy [1], [3], [8]. The dissipation itself scales as $|w_{mn}|^2$ with the beamforming state, so a consistent coupling must preserve component identity, apply element-dependent power scaling, and accommodate non-radiating parts absent from the periodic EM model.

II. Proposed Automated HFSS-to-Icepak Methodology

Main steps of the proposed workflow are illustrated in Fig. 2. The unit cell is first characterized in HFSS to obtain its EM response and the per-object dissipated powers (step 1), while the TRM is characterized in HFSS or HFSS 3D Layout over temperature, yielding its phase deviation (step 2). Unit-cell and power-amplifier losses are scaled by the tapering amplitude A_{mn} and applied to the full-array Icepak model (step 3), from which the per-cell unit-cell thermal fingerprint $\boldsymbol{\Theta}_i$ and the per-cell TRM temperature T_{mn}^{TRM} are extracted. The TRM temperatures are mapped onto the TRM phase–temperature curve to obtain the element-wise phase errors (step 4). In parallel, the unit cells are grouped into clusters based on their object-level thermal state. For each cluster ℓ and each physical object p, a representative temperature $\overline{T}_{\ell,p}$ is computed from the per-object, per-cell mean temperatures within the cluster, and a dedicated material instance evaluated at $\overline{T}_{\ell,p}$ is assigned to object p in all cells of cluster ℓ (step 5). This reduces the array to a small number of electromagnetically distinct unit-cell variants, making the full-array simulation tractable via FADDM in HFSS while preserving the per-object thermal information up to the clustering approximation. Finally, the clustered cells, the tapering and the phase errors are combined in the full-array FADDM model (step 6), yielding thermal-informed array performance.

A. Electromagnetic Loss Extraction

The reference unit cell is solved in HFSS for the prescribed excitation and operating frequency. HFSS provides the volumetric loss density $q_{\mathrm{v}}(\mathbf{r})$, corresponding to `VolumeLossDensity`, and the surface loss density $q_{\mathrm{s}}(\mathbf{r})$, corresponding to `SurfaceLossDensity`.

The dissipated power associated with a solid domain V_p or a surface S_p is calculated as

$$P_{\mathrm{v},p} = \int_{V_p} q_{\mathrm{v}}(\mathbf{r})\,\mathrm{d}V, \tag{2}$$

$$P_{\mathrm{s},p} = \int_{S_p} q_{\mathrm{s}}(\mathbf{r})\,\mathrm{d}S. \tag{3}$$

The appropriate integral is selected according to the electromagnetic representation assigned to each object. The extracted power remains associated with the corresponding component identifier to support its subsequent assignment in the thermal model.

B. Array-Level Power Scaling

The losses extracted from the reference unit cell are scaled according to the magnitude of the excitation assigned to each array element. For an excitation weight w_{mn}, the dissipated power associated with the p-th object of the (m,n)-th cell is defined as

$$P_{p,mn} = |w_{mn}|^2\, P_p^{\mathrm{cell}}, \tag{4}$$

where P_p^{cell} is the corresponding loss extracted from the reference unit cell at its nominal source-power setting (chosen so that $|w_{mn}|^2$ carries the physical per-cell power scaling). Eq. (4) is an *infinite-array extended by taper* approximation: it assumes that the local field distribution in each array cell equals the periodic unit-cell field scaled by w_{mn}, neglecting finite-array edge effects and scan-induced coupling variations. The approximation is accurate for large arrays scanned near the characterization angle; it degrades for heavily tapered edge cells and large scan angles. This formulation allows the same reference solution to be mapped to different

array sizes, beamforming states, and magnitude tapers without repeating the complete electromagnetic analysis.

C. Automated Icepak Model Construction

The aperture-level geometry is constructed in Icepak from the reference cell and the additional system-level components required by the thermal analysis. The automation procedure creates the array layout, preserves the component hierarchy, assigns materials, and generates the thermal sources corresponding to the scaled electromagnetic losses.

The mapped losses are assigned as surface or volumetric heat sources according to their original electromagnetic representation.

Components not represented in the electromagnetic unit-cell model, including TRMs, supports, enclosures, thermal spreaders, and cooling structures, can be included independently according to the required thermal-modeling fidelity. The same freedom applies to the thermal boundary conditions: solar irradiance, ambient conditions, convective and radiative exchanges can be added independently of the EM model, matching the required thermal-modelling fidelity.

D. Object-Level Thermal Post-Processing

After the Icepak solution, the temperature field is automatically processed for each array cell. Rather than reducing the complete cell to a single temperature value, thermal statistics are extracted separately for the relevant physical objects composing the cell, such as the radiating element, substrate, feed network, and active components.

For object p of cell c_i, the thermal state is described by its weighted mean temperature $\overline{T}_{i,p}$, temperature dispersion $\sigma_{T,i,p}$, and minimum and maximum temperatures. Surface or volume weighting is adopted according to the physical representation of the object.

The resulting object-level data provide a thermal fingerprint of each unit cell,

$$\boldsymbol{\Theta}_i = \left\{ \overline{T}_{i,p}, \sigma_{T,i,p}, T_{i,p}^{\min}, T_{i,p}^{\max} \right\}_{p=1}^{N_o}. \tag{5}$$

Here N_o is the number of physical objects in the cell, assumed identical across cells sharing the same unit-cell representation (enforced by the group partitioning of Sec. II-E). The extracted thermal fingerprints are arranged according to the aperture coordinates for hotspot assessment and constitute the input to the clustering stage.

E. Object-Aware Thermal Clustering for Temperature Backannotation

Directly assigning an independent thermal state to every array cell would require a distinct temperature-dependent electromagnetic representation for each cell, effectively eliminating the computational benefits of the finite array domain decomposition method. An object-aware clustering stage is therefore introduced to preserve the dominant thermal nonuniformities while limiting the number of electromagnetic thermal states.

Before clustering, cells are divided into groups according to the array component represented by their unit-cell model, such as passive corner cells, passive side cells, or different types of active cells. Clustering is performed independently within each group, ensuring that only cells sharing the same unit-cell representation and the same set of corresponding physical objects can belong to the same thermal cluster.

For each cell c_i, the object-level thermal statistics are condensed into a two-component feature vector,

$$\mathbf{x}_i = \begin{bmatrix} \overline{T}_i^{\mathrm{met}} \\ \overline{T}_i^{\mathrm{nmet}} \end{bmatrix}, \tag{6}$$

where $\overline{T}_i^{\mathrm{met}}$ and $\overline{T}_i^{\mathrm{nmet}}$ are the mean temperatures of the metallic and non-metallic objects within cell c_i. This two-component representation separates conductive and dielectric regions, which affect the EM response differently.

Within each unit-cell group, the cells are partitioned by minimizing the within-cluster thermal dispersion,

$$\mathcal{J}_g(K_g) = \sum_{\ell \in \mathcal{L}_g} \sum_{c_i \in \mathcal{S}_\ell} \left\| \mathbf{W}_g \widetilde{\mathbf{x}}_i - \boldsymbol{\mu}_\ell \right\|_2^2, \tag{7}$$

where $\widetilde{\mathbf{x}}_i$ is the standardized feature vector, $\mathbf{W}_g = \mathrm{diag}(w_1, w_2)$ contains the feature weights (equal weights $w_1 = w_2 = 1$ in this work), $\mathcal{S}_\ell$ is the ℓ-th cluster, and $\boldsymbol{\mu}_\ell$ its centroid. Here ℓ is the global cluster index and $\mathcal{L}_g \subset \{1, \ldots, K_{\mathrm{tot}}\}$ is the subset of clusters assigned to group g, with $|\mathcal{L}_g| = K_g$ and $\bigcup_g \mathcal{L}_g = \{1, \ldots, K_{\mathrm{tot}}\}$.

The engineer specifies a total cluster budget K_{tot} (with $K_{\mathrm{tot}} \geq N_g$, where N_g is the number of unit-cell groups). Each group receives one cluster initially; the remaining $K_{\mathrm{tot}} - N_g$ clusters are allocated greedily by maximizing the absolute inertia drop

$$G_g(K_g) = \mathcal{J}_g(K_g) - \mathcal{J}_g(K_g + 1), \tag{8}$$

so that each additional cluster is spent where it most reduces the within-group thermal dispersion. Because $\mathcal{J}_g$ scales with the group size, this criterion naturally favors groups that combine a large population with high internal variability—typically the active cells, which both radiate and span the widest temperature range—over small groups with narrow spread (e.g. passive corner or edge cells).

For each resulting cluster, a representative temperature is obtained per physical object by volume-weighted averaging across the cells of the cluster,

$$\overline{T}_{\ell,p} = \frac{\sum_{c_i \in \mathcal{S}_\ell} V_{i,p} \, \overline{T}_{i,p}}{\sum_{c_i \in \mathcal{S}_\ell} V_{i,p}}, \tag{9}$$

where p runs over the N_o physical objects shared by all cells of cluster ℓ (enforced by the group partitioning of Sec. II-E). All cells in the cluster share these N_o representative temperatures in the EM model, one per

object, rather than a single temperature per material class.

The representative states are then automatically back-annotated to the FADDM model: for each cluster ℓ, a variant of the 3D component is instantiated in which every object p receives its material-property vector $\mathbf{p}(\overline{T}_{\ell,p})$ evaluated at the cluster temperature. Each array cell (m,n) is then mapped to the variant of its assigned cluster c_{mn}, and its excitation is updated to $A_{mn}e^{j(\phi_{mn}+\Delta\phi_{mn})}$. The resulting finite array uses at most K_{tot} distinct cell definitions and exposes the thermal non-uniformity directly in the EM response.

III. Test Case Setup

A. HFSS, Unit Cells

The electromagnetic test case was developed in Ansys HFSS using a 3D Component Finite Array Domain Decomposition Method (FADDM) formulation. Each radiating cell was represented by a parameterized 3D component, allowing the finite array to be assembled while retaining the individual electromagnetic description of its constituent elements. The complex excitation of the (m,n)-th element was defined as $w_{mn} = A_{mn}\exp(j\phi_{mn})$, with A_{mn} following a Taylor amplitude taper designed for a $-25\,\mathrm{dB}$ sidelobe level and ϕ_{mn} following the broadside beam-steering law ($\phi^0_{mn} = 0$). The source-power setting of the reference unit cell was fixed so that the most strongly excited cell receives $+5\,\mathrm{W}$, while the remaining cells scale proportionally to $|w_{mn}|^2$. The 3D Component FADDM solution provides the electromagnetic field distribution, the mutual coupling among the array elements, and the spatially resolved dielectric and conductive losses required for the subsequent thermal analysis.

B. HFSS 3D Layout, TRM

The transmit/receive module was characterized separately using an HFSS 3D Layout model and swept over temperature to extract the phase deviation, $\Delta\phi(T)$, at the operating frequency (Fig. 3). The PA contribution is included indirectly through its dissipated power, which is modeled as a heat source in the thermal analysis and drives the temperature distribution across the TRM. The resulting thermally induced phase deviations are subsequently back-annotated into the full-array model. Temperature-dependent phase distortions originating from the PA electrical behavior itself are not considered in the present work. A more comprehensive electro-thermal characterization of the PA is left for future investigation.

C. Icepak

In the proposed workflow, Icepak assembles the complete system, from the radiating unit cells and the TRMs up to any desired level of integration. The practical limit

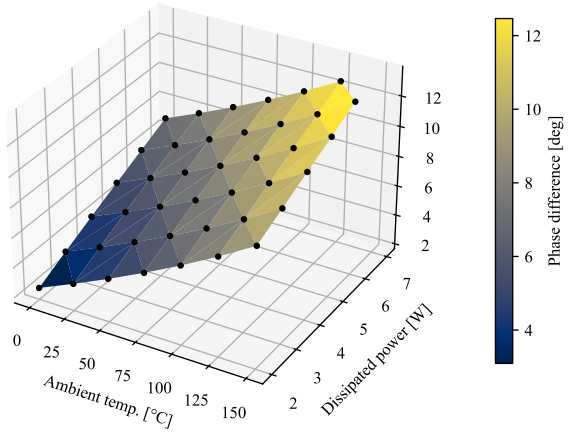


Fig. 3: Phase difference of the TRM output signal versus ambient temperature and PA dissipated power.

is computational cost, driven mainly by the number of interfaces between geometries. The model considered in this work includes the following thermal loads and boundary conditions:

- radiative exchange with deep space, modeled as a black-body environment at $T_\infty = +3\,\mathrm{K}$;
- solar irradiance at normal incidence on the radome;
- RF losses in each unit cell, computed in HFSS and applied per object after the taper scaling of Eq. (4);
- heat dissipation in the TRM passive components (LNA, circulator, core chip, limiter) with the same taper scaling;
- PA thermal dissipation: each power amplifier is represented by a heat source whose reference value is derived from nominal datasheet data and whose spatial distribution across the array is modulated by the excitation magnitude $|w_{mn}|^2$ assigned to the corresponding cell.

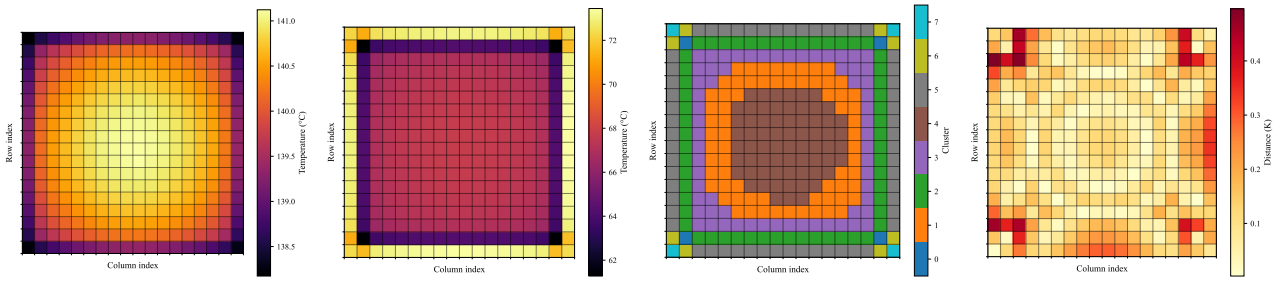
Energy equation. Every object is assigned its own thermal conductivity κ_p (one value per object p). The vacuum enclosure is represented by a medium with negligible conductivity ($\kappa_{\mathrm{vac}} = +10^{-5}\,\mathrm{W\,m^{-1}\,K^{-1}}$), so no convective transport occurs. Under steady-state conditions the energy equation reduces to

$$\nabla\cdot(\kappa_p\nabla T) + \dot{q}_p = 0 \quad \text{in } \Omega_p, \tag{10}$$

where the volumetric heat source $\dot{q}_p$ is the HFSS loss density q_{v} of object p in the reference unit cell, scaled per cell by $|w_{mn}|^2$ [Eq. (4)]. Surface losses q_{s} are applied analogously as boundary heat fluxes.

Radiation model. Thermal radiation is modeled using the discrete ordinates method to account for radiative heat exchange among the unit-cell metallization, the radome, and deep space. Vacuum is assumed to be a non-participating medium.

Boundary conditions. The outer radome absorbs incident solar radiation according to $\alpha_s G_s$, where $G_s \approx$



(a) Metallic $\overline{T}$. (b) Non-metallic $\overline{T}$. (c) Cluster assignment. (d) Distance d_i to centroid.

Fig. 4: Object-aware thermal clustering of the 18×18 array: (a) volume-weighted mean temperature of the metallic objects; (b) same for the non-metallic (radome) objects; (c) greedy cluster assignment with $K_{\mathrm{tot}} = 8$; (d) per-cell distance from the cluster representative state.

$+1361\,\mathrm{W\,m^{-2}}$, and rejects heat by thermal radiation toward deep space, modeled as a sink at $T_\infty = +3\,\mathrm{K}$. The cooling system is not modeled explicitly. Instead, its effect is approximated by an equivalent heat-sink boundary condition applied to the bottom surface of the cold plate, with an effective heat-transfer coefficient of $h = +85\,\mathrm{W\,m^{-2}\,K^{-1}}$. This value was selected to reproduce the thermal removal capability of the cooling subsystem while maintaining the active devices within their specified operating temperature range.

IV. Results and Discussion

Clustering.

The 18×18 array under test comprises three unit-cell types: four passive corner cells, 64 passive side cells, and 256 active cells. The object-aware clustering algorithm was applied with a total budget of $K_{\mathrm{tot}} = 8$ clusters and equal feature weights $w_1 = w_2 = 1$; the greedy allocation of Eq. (8) spends the clusters where the within-group inertia drop is largest, naturally concentrating them on the active group that combines a large population with the widest thermal spread. The volume-weighted mean temperature of the metallic objects (Fig. 4a) spans 1.8 K on the active cells, 0.8 K on the side cells, and 0.1 K on the corner cells, while the non-metallic radome objects (Fig. 4b) show a reversed spatial distribution, warmer along the passive frame and colder above the active aperture. This opposite behavior motivates the two-component feature vector of Eq. (6).

As a consequence (Fig. 4c), the greedy allocation driven by Eq. (8) assigns one cluster to the passive corner group, two to the passive side group, and the remaining five to the active group, saturating the budget where additional thermal states matter most.

The quality of the reduced representation is quantified by the distance $d_i = \|\mathbf{W}_{g(i)}(\mathbf{x}_i - \boldsymbol{\mu}_{\ell(i)})\|_2/\sqrt{N_f}$ (in kelvin, with $N_f = 2$ the number of features, $g(i)$ the group of cell c_i, and $\ell(i)$ its assigned cluster) between each cell and the centroid of its assigned cluster. Fig. 4d shows that the residuals (mean 0.13 K, maximum 0.50 K) concentrate in the transition zones between active aperture and passive frame; they are more than one order of magnitude smaller than the overall thermal excursion (tens of kelvin), confirming that the eight representative states retain the dominant thermal nonuniformities for the back-annotation to the FADDM solver.

The approximate asymmetries visible in Fig. 4c are a finite-volume solver artifact, comparable to the half-distance between neighbouring centroids, and do not affect the back-annotation.

Thermal-informed array performance. At $+6.8\,\mathrm{GHz}$ in the $\phi = +0^\circ$ cut, the temperature-aware array reaches a peak gain of $+27.80\,\mathrm{dB}$ against $+28.69\,\mathrm{dB}$ of the reference, yielding

$$\Delta G_{\max} = G_{\max,\mathrm{th}} - G_{\max,\mathrm{ref}} = -0.89\,\mathrm{dB}. \qquad (11)$$

The difference does not come from an externally imposed statistical perturbation but from the temperature field reassigned to the constituent objects of the HFSS 3D components (Fig. 5).

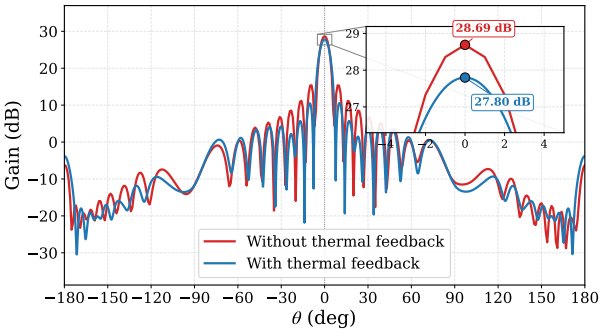


Fig. 5: Co-polar radiation pattern at $+6.8\,\mathrm{GHz}$, $\phi = +0^\circ$ cut: reference HFSS solution versus temperature-aware solution obtained after mapping the Icepak temperature field back to the array.

The main-beam detail (Fig. 6) shows the half-power beamwidth broadening from $+5.79^\circ$ to $+6.48^\circ$ ($\Delta\mathrm{HPBW} \approx +0.69^\circ$) consistent with the loss of aperture coherence. The sidelobe structure and null distribution are substantially preserved and no significant beam-pointing displacement is observed. This is consistent with a temperature-induced perturbation that is nearly

symmetric across the aperture and whose dominant effect is a reduction of aperture coherence (gain and HPBW) rather than a linear phase tilt, which would steer the beam.

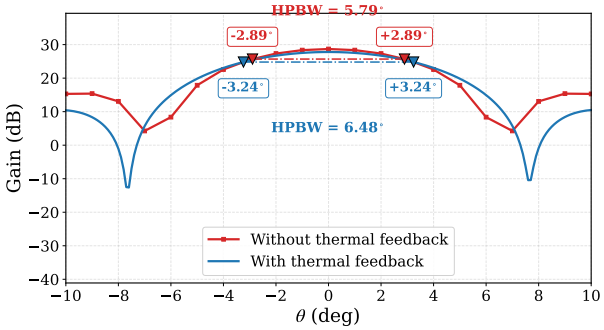


Fig. 6: Detail of the main-beam region, highlighting the gain reduction and the half-power beamwidth.

Fig. 7 isolates the phase-error contribution by applying only the TRM phase deviation $\Delta\phi(T)$: the pattern peak shifts from $+0°$ to $+0.20°$ ($\Delta\theta = +0.20°$), confirming a small but measurable beam-pointing displacement that is masked by the dominant amplitude reduction when both effects are coupled.

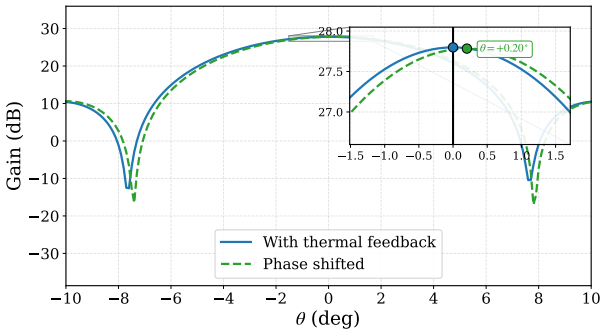


Fig. 7: Pattern comparison when only the phase shift induced by the TRM temperature is applied, isolating the phase-error contribution from the material-property update.

V. Conclusions

An automated workflow developed in Ansys Electronics Desktop connecting HFSS and Icepak for the electro-thermal analysis of phased-array antennas has been presented and demonstrated on an 18×18 active aperture. The scripted mapping transfers object-level EM losses to the thermal model with proper taper scaling, runs the aperture-level thermal solution, and feeds the resulting temperature field back to HFSS through an object-aware clustering stage that preserves the dominant thermal nonuniformities while limiting the number of distinct electromagnetic states.

For the test case, the thermal-informed solution shows a reduction in peak gain and a broadening of the main beam when only material-property feedback is applied, while isolating the TRM phase-shift feedback introduces measurable beam-pointing shift—two complementary mechanisms that the workflow captures separately.

Because HFSS and Icepak geometries are decoupled, the workflow extends naturally to richer thermal scenarios. Its main value is removing the manual geometry, loss-transfer, source-assignment, and back-annotation steps that today prevent electro-thermal analysis from being used routinely in early-phase AESA design.

Future work should also investigate the role of in-aperture thermal gradients, which experimental evidence in the literature [1] shows can significantly affect gain, sidelobe level, and pointing accuracy.

References

[1] Y. Wang, C. Wang, P. Lian, S. Xue, J. Liu, W. Gao, Y. Shi, Z. Wang, K. Yu, X. Peng, B. Du, and S. Xiao, "Effect of Temperature on Electromagnetic Performance of Active Phased Array Antenna," *Electronics*, vol. 9, no. 8, Art. no. 1211, 2020, doi: 10.3390/electronics9081211.

[2] F. T. Celik, C. Fager, A. Yarovoy, and Y. Aslan, "An Iterative Electro-Thermal Model for an Active Antenna Element and its Application to Arrays," in *Proc. 2025 55th European Microwave Conference (EuMC)*, 2025, pp. 1127-1130, doi: 10.23919/EuMC65286.2025.11235236.

[3] Y. Wang, P. Xian, Q. Guo, Y. Qin, S. Xue, P. Lian, L. Zhang, Z. Wang, W. Wu, and C. Wang, "Electromagnetic-Thermal Coupling and Optimization Compensation for Missile-Borne Active Phased Array Antenna," *Technologies*, vol. 14, no. 1, Art. no. 67, 2026, doi: 10.3390/technologies14010067.

[4] Y. Aslan, J. Puskely, J. H. J. Janssen, M. Geurts, A. G. Roederer, and A. Yarovoy, "Thermal-Aware Synthesis of 5G Base Station Antenna Arrays: An Overview and a Sparsity-Based Approach," *IEEE Access*, vol. 6, pp. 58868-58882, 2018, doi: 10.1109/ACCESS.2018.2873977.

[5] H. H. Zhang, X. Y. Liu, Y. Liu, Z. C. Fan, and H. L. Du, "Thermal-Mechanical-Electromagnetic Multiphysics Simulation of Satellite Phased Array Antenna Based on DGTD and FEM Method," *IEEE Journal on Multiscale and Multiphysics Computational Techniques*, vol. 9, pp. 236-246, 2024, doi: 10.1109/JMMCT.2024.3428517.

[6] H. Liu, W. Wang, D. Tang, L. Zhang, Y. Wang, and E. Miao, "Thermal Deformation Modeling for Phased Array Antenna Compensation Control," *Sensors*, vol. 22, no. 6, Art. no. 2325, 2022, doi: 10.3390/s22062325.

[7] Z. Liao, S. Ma, S. Li, X. Wang, and T. Wan, "Electromagnetic-Thermal-Stress Modeling of Antenna Array Based on Finite Element Domain Decomposition Method," *IEEE Antennas and Wireless Propagation Letters*, vol. 23, no. 12, pp. 4323-4327, 2024, doi: 10.1109/LAWP.2024.3445408.

[8] A. D. Brown, *Active Electronically Scanned Arrays: Fundamentals and Applications*. Hoboken, NJ, USA: Wiley-IEEE Press, 2021.